

Veterinary / Medical Management System

Complete Feature Analysis & Gap Report

1. Existing Features

1.1 Billing / Sales

- Customer name autocomplete from existing customers
- Phone number entry with 10-digit validation
- Previous due amount auto-fetched on customer selection
- Doctor name selection from registered doctors (autocomplete)
- Doctor phone auto-filled on doctor selection
- Two-step medicine combo: search by name then select batch/expiry
- Schedule validation: scheduled medicines require doctor name
- Expiry check: blocks billing of expired medicines
- Stock check: blocks billing if quantity exceeds available stock
- Type-aware rate calculation (tablets: MRP / tablets-per-strip)
- Per-medicine discount percentage
- Overall bill discount percentage with auto rounding
- Three payment modes: Cash, Online, Phone Pay
- Due amount and total due with previous due carry-forward
- Credit amount tracking when customer overpays
- Bill cleared / Account cleared flags
- Bill number auto-generated (BILL + timestamp)
- Bill preview in browser (HTML print), Ctrl+P to reprint, F5 to generate
- Bill edit and delete from Sales History (delete restores stock)
- GST percentage display (included in MRP)

1.2 Purchase Management

- Supplier autocomplete with auto-fill address/phone/GSTIN/DL
- Purchase date, supplier bill number entry
- GST calculation: Discount Before GST or Discount After GST
- Medicine autocomplete with auto-fill type/manufacturer/HSN/GST/MRP/rate/schedule
- Medicine types: Tablet, Syrup, Injection, Vial, Ointment, Powder, Bolus, Liquid, Liniment, Gel, Vaccine, Granules
- Tablet/Bolus: Stripes qty + Pack Size (e.g. 6'S) + Free stripes
- Other types: Pack Size (e.g. 120ml) + Units qty + Free units
- HSN code, GST%, MRP, Rate, Manufacturer, Batch No, Expiry (MM/YY), Schedule, Content/Drug
- Per-item discount, overall discount (rupee), rounding, CGST/SGST display
- Previous due and credit auto-loaded from last purchase
- Stock auto-updated on save (tablets: stripes x pack size + free)
- Purchase number auto-generated (PUR + timestamp)
- Purchase edit and delete from Purchase History (delete reduces stock)
- Bill cleared / Account cleared flags per supplier
- Web Purchase Entry: browser-based fast entry with JSON export
- Import Purchases: paste/load JSON, review each bill, submit all

1.3 Inventory Management

- Full medicine list: Name, Type, Batch, Expiry, Stock, Unit (Pack Size), MRP, Rate, Manufacturer, Schedule
- Unit column shows 6'S for tablets/bolus, raw value (120ml, 1kg) for others
- Search by name or batch (starts-with priority), filter by Type/Stock Status/Expiry/Schedule
- Color-coded rows: red = out of stock/expired, orange = low stock/near expiry

- Low stock and near expiry thresholds configurable per medicine type
- Edit medicine (all fields), View details (purchase + sales history per medicine)
- Delete medicine (blocked if has sales history)
- Optional shelf/location column (rack-section-box system)
- Summary: total medicines, low stock, out of stock, total value (stock x MRP)

1.4 Sales History

- Full sales list with date, customer, phone, amounts, payment breakdown, due/credit
- Filter by date range, customer, due status, schedule
- Color-coded: green = account cleared, yellow = bill cleared, red = has due
- Summary: total bills, total sales, total due, total profit, today/month revenue/profit, payment mode breakdown, total discount
- View bill details, Edit bill, Print bill (HTML), Delete bill
- Export Schedule Report: A4 HTML report filterable by schedule and date range

1.5 Purchase History

- Full purchase list: bill no, date, supplier, total, due, credit, net due, item count
- Filter by date range, supplier, due status, schedule
- Color-coded: green = account cleared, yellow = bill cleared, red = has due
- Summary: total purchases, total amount, total due, total credit, total items, average purchase
- Edit purchase (full re-entry), Delete purchase (reduces stock)

1.6 Customer Management

- Customer list with name, phone, address, document name
- Search customers, view customer sales history
- Customer due and credit tracking across all bills

1.7 Settings & Configuration

- Pharmacy profile: name, address, phone, email, GSTIN, DL number, GST enabled toggle
- Doctor management: add/edit/delete with name, registration number, phone
- Supplier management: add/edit/delete with name, address, phone, GSTIN, DL numbers
- Shelf management: rack / section / box hierarchy with location assignment
- 8 themes (dark/light), font size (7-20), row count per page (all modules)
- Unit/measure and default quantity per medicine type
- Schedule list and medicine type list: add/remove
- Low stock and near expiry thresholds per medicine type
- Export data: Sales CSV, Purchases CSV, Inventory CSV, Export All
- Delete all data (password protected), Import purchases via JSON
- Keyboard shortcuts reference page

1.8 UI / UX Features

- Modern dark/light themes via ttkbootstrap
- Full keyboard navigation: arrow keys, Enter, Escape, Tab across all pages
- Number keys 1-7 for instant page navigation, F2/F5/F6/Ctrl+P shortcuts
- Searchable combo boxes, two-step medicine combo for billing
- Auto-scroll to focused widget, persistent billing/purchase pages
- Right-click context menus on all lists
- Custom font (Nirmala UI) with configurable size
- Single-file EXE build (PyInstaller), Web purchase entry (browser-based)

2. Missing Features (Gaps vs Marg / Full Medical Software)

2.1 Critical — GST & Compliance

- GSTR-1 report generation (outward supplies)
- GSTR-3B report generation (monthly summary)
- GST invoice with proper CGST/SGST/IGST breakup on bill printout
- HSN-wise GST summary report
- E-way bill generation
- Tax invoice vs retail invoice toggle
- GST rate master (medicine-wise GST slab management)

2.2 Critical — Returns

- Sales return / credit note: customer returns medicine, stock restored, amount credited
- Purchase return / debit note: return medicine to supplier, stock reduced, amount debited
- Partial return support (return some items from a bill)
- Return history report
- Expired medicine return to supplier tracking

2.3 Critical — Accounting & Ledger

- Customer ledger: full account statement per customer (all bills, payments, dues)
- Supplier ledger: full account statement per supplier (all purchases, payments, dues)
- Day book / cash book: daily transaction summary
- Profit & loss statement
- Balance sheet
- Bank/cheque payment tracking for suppliers
- Payment receipt generation
- Outstanding dues report (all customers with pending dues)
- Outstanding payables report (all suppliers with pending dues)

2.4 Important — Inventory

- Opening stock entry (for migrating from another system)
- Stock adjustment / manual correction entry
- Reorder level per medicine with auto-alert when stock falls below
- Purchase order generation to supplier
- Barcode scanning support for medicine selection
- Batch-wise and expiry-wise stock reports
- Dead stock report (medicines with zero movement)
- Stock valuation report (at purchase rate and MRP)
- Medicine master with standard drug database / generic name

2.5 Important — Billing

- Bill number series configuration (custom prefix, auto-increment)
- Hold bill / save draft and resume later
- Quotation / estimate generation
- Wholesale vs retail price toggle per customer
- Free item scheme (buy X get Y free)
- Label printing for medicine bottles/strips
- Thermal printer / POS printer support
- Bill reprint with original bill number

2.6 Important — Reports

- Medicine-wise sales report (how much of each medicine sold)
- Doctor-wise prescription report
- Customer-wise and supplier-wise purchase report
- Monthly/yearly sales comparison
- Top selling and slow moving medicines report
- Margin/profit per medicine report
- GST summary report (CGST + SGST collected)
- Day-end closing report
- PDF export for all reports (currently only HTML/CSV)

2.7 Missing — Security & Multi-User

- User login with username and password
- Role-based access control (admin, billing staff, purchase staff)
- Audit trail: who created/edited/deleted each record
- Multi-user / network mode (currently single-user SQLite only)

2.8 Missing — Data Safety

- Automatic database backup (daily/weekly to a folder)
- Backup to cloud / Google Drive
- Restore from backup
- Transaction log / undo history

2.9 Minor Missing

- SMS / WhatsApp due reminders to customers
- Email bill to customer
- Dashboard with charts (sales trend, stock alerts, due summary)
- Prescription image attachment
- Multiple branch / store support
- Customer loyalty points / reward system

3. Can This Be Used for Normal Medical / Pharmacy Shops?

Yes — with minor changes:

- The core billing, purchase, inventory, GST, and due management logic is identical for both veterinary and human pharmacies
- Schedule validation (H, H1, X, G, etc.) is standard for both
- The only veterinary-specific element is the app name — the business logic is standard pharmacy logic
- Change "Veterinary Management System" to your shop name in main.py (title and welcome label)
- Change the pharmacy profile in Settings to your shop details
- Medicine types (Tablet, Syrup, Injection, etc.) are the same for both
- DL number, GSTIN, schedule validation — all standard pharmacy requirements already implemented

4. Overall Assessment Summary

Module / Area	Status	Notes
Basic Billing	Complete	Full billing with due, discount, multi-payment
Purchase Entry	Complete	Full purchase with GST, batch, expiry, free qty
Inventory	Good	Missing: opening stock, reorder alerts, barcode

Sales History	Good	Missing: medicine-wise report, PDF export
Purchase History	Good	Missing: supplier ledger, payment receipts
Customer Management	Basic	Missing: full ledger, SMS alerts
GST Compliance	Partial	Tracks GST but no GSTR-1/3B reports
Returns	Missing	No sales return or purchase return module
Accounting / Ledger	Missing	No day book, P&L, balance sheet
Security / Login	Missing	No user login or access control
Backup / Restore	Missing	No auto backup or restore
Reports	Basic	Only CSV export and schedule HTML report
Normal Medical Use	Yes	Works for human pharmacy with name change only

Bottom Line: This is a solid, well-built basic pharmacy management system suitable for small veterinary clinics and medical shops. It covers core daily operations (billing, purchase, inventory, history) very well. For a full Marg-level replacement, the critical additions needed are: GST returns (GSTR-1/3B), sales/purchase returns, customer/supplier ledger, user login, and automatic backup.